

Multi-AGV Warehouse Simulator

Fleet Sizing, Dispatching, and Collision-Aware Path Planning

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August 2026

Abstract

This project develops and evaluates a multi-AGV warehouse simulator that combines task generation, task dispatching, collision-aware path planning, and reproducible performance experiments. Robot movement is coordinated using Space-Time A* and a reservation table, while task assignment is evaluated using nearest-available and first-available dispatching. The experimental study contains 180 headless runs covering six fleet sizes, three task-generation intervals, two dispatch strategies, and five random seeds per configuration.

The results show a clear capacity threshold. A single AGV is limited to about 2.12 tasks/min regardless of workload. Under the highest workload, three AGVs raise throughput to 6.04 tasks/min and completion within the 600-second horizon to 95.88%, while a fourth AGV mainly reduces queueing rather than increasing throughput. The medium workload requires approximately two to three AGVs, whereas two AGVs are sufficient for the lowest workload. Under the 10-second workload, nearest-available dispatching reduces total travel distance by about 7-13% for fleets of four to six AGVs compared with first-available dispatching, while the cycle-time improvement is smaller. These results indicate that fleet sizing is the dominant factor for capacity and delay, whereas dispatch policy becomes more important for movement efficiency once spare fleet capacity is available.

Keywords: automated guided vehicles, warehouse simulation, Space-Time A*, multi-agent pathfinding, dispatching, fleet sizing

1. Introduction

Automated Guided Vehicles (AGVs) are widely used in warehouses and manufacturing environments to transport materials between different locations. When several AGVs operate in the same environment, the system must assign transport tasks efficiently and coordinate robot movement so that collisions and unnecessary delays are avoided.

This project presents a simulation of an intelligent warehouse containing multiple autonomous AGVs. Transport tasks are generated during the simulation and assigned to available robots. Each task consists of travelling to a pickup location and then transporting the item to a delivery location.

Robot routes are calculated using path-planning algorithms. Because several robots share the same warehouse, route planning must also consider the position and planned movement of other AGVs. The simulator therefore uses Space-Time A* together with a reservation mechanism to coordinate robot paths and prevent conflicts.

The simulator is also used as an experimental platform. Different AGV fleet sizes and task-generation rates are tested to determine how they affect warehouse performance. In addition, two task-assignment strategies are compared: nearest-available and first-available.

The engineering questions addressed here are consistent with established AGV research, where fleet sizing, routing, dispatching, idle-vehicle positioning, and deadlock avoidance are treated as interacting control problems. Large-scale robotic warehouse systems also demonstrate that coordination and traffic management become increasingly important as the number of autonomous vehicles grows. This project studies a smaller, controlled version of those problems so that fleet capacity and dispatching effects can be measured directly.

2. Research Questions and Contributions

The main objective of this project is to design and evaluate a multi-AGV warehouse simulation in which transport tasks can be assigned and completed safely and efficiently.

The project focuses on two main research questions:

1. How do AGV fleet size and task arrival rate affect warehouse performance?
2. Does assigning a task to the nearest available AGV improve system efficiency compared with assigning it to the first available AGV?

To answer these questions, the simulator records performance metrics such as throughput, average queue time, average cycle time, fleet utilization, completion rate, travel distance, and drain time.

The main contributions of the project are: (1) an integrated multi-AGV simulator with coordinated collision-aware routing; (2) a reproducible experiment design covering 180 runs across fleet size, workload, dispatch strategy, and random seed; and (3) a quantitative comparison that links throughput, delay, utilization, completion, draining, and travel distance rather than relying on a single performance measure.

3. System Design

3.1 Warehouse Environment

The warehouse environment is represented as a two-dimensional grid. The default warehouse has 15 rows and 20 columns. Each grid cell represents one position that can either be free space, a shelf, a pickup location, or a delivery location.

Shelves are treated as static obstacles and cannot be entered by an AGV. Free cells, pickup locations, and delivery locations are traversable. Before a robot moves into a cell, the system checks whether the position lies within the warehouse boundaries and whether it is not occupied by a shelf.

Robot movement is restricted to four directions: up, down, left, and right. Diagonal movement is not permitted. For a given robot position, the warehouse therefore generates neighbouring cells in these four directions and removes any positions that are outside the warehouse or blocked by shelves.

The default layout contains four vertical groups of shelves, creating aisles through which the AGVs can travel. Two pickup locations and two delivery locations are defined in the environment. Transport tasks select from these locations, requiring robots to navigate through the warehouse between pickup and delivery points.

Figure 1 presents the graphical interface of the implemented warehouse simulator. It shows the warehouse grid, shelf layout, pickup and delivery locations, positions of multiple AGVs, and a status panel containing information about the current state of each robot.

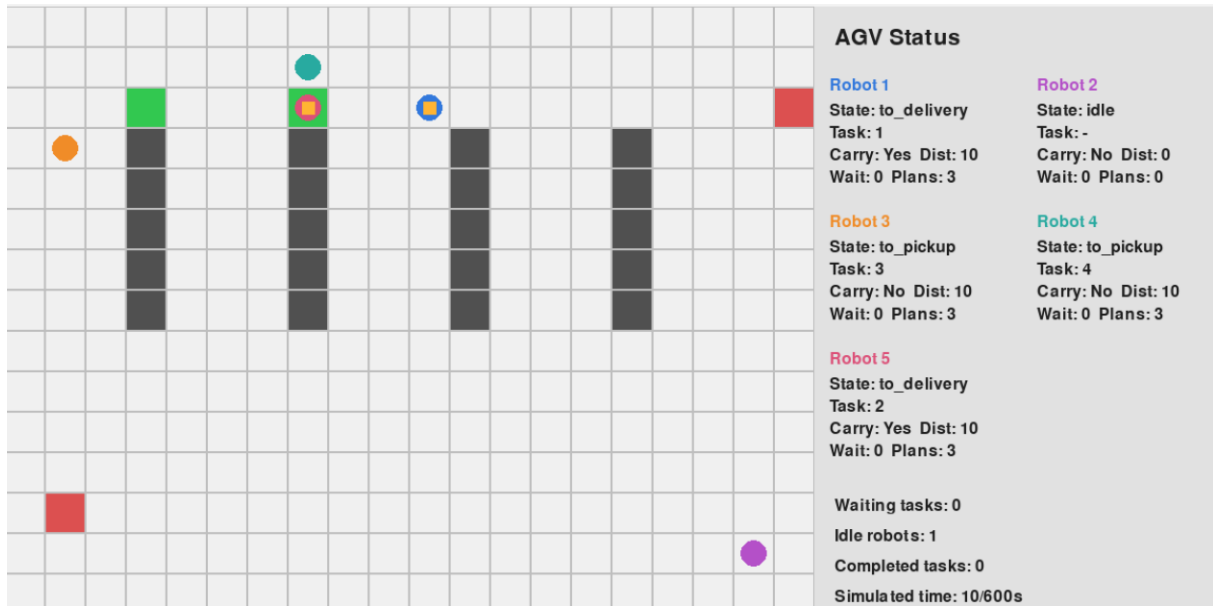


Figure 1. Visualization of the warehouse environment with five AGVs, shelf obstacles, pickup and delivery locations, and the AGV status panel.

3.2 AGV Robot Model

Each AGV is represented as an individual robot object with its own identifier, position, home position, assigned path, current task, and operating state. The robot's initial position is also stored as its home position, which allows the AGV to return to its starting location after completing a transport task. Each robot is assigned a different colour so that individual AGVs can be distinguished in the graphical simulation.

The behaviour of an AGV is described using several operating states. A robot begins in the **idle** state. When a transport task is assigned, its state changes to **to_pickup**, and the robot travels towards the required pickup location. After reaching the pickup point, the transported item is collected and the robot proceeds towards the delivery location in the **to_delivery** state. After completing the delivery, the robot can return to its home position before becoming idle again.

Each robot follows a path represented as a sequence of grid positions. The robot advances through this sequence one position at a time. A validation check ensures that a newly assigned path begins at the robot's current position, helping detect incorrect routes before they are executed.

The movement model also supports intentional waiting. Space-Time A* can produce a path in which two consecutive positions are identical, meaning that the AGV remains in the same grid cell for one simulation step instead of moving. This allows robots to wait when necessary to avoid conflicts with other AGVs. Actual movement increases the robot's travelled-distance counter, while task-related waiting increases its waiting-time counter. When a robot is completely idle, its idle-time counter is increased instead.

Several statistics are maintained separately for each AGV, including total distance travelled, task-related waiting steps, idle steps, and the number of routes created. These robot-level measurements are later combined to calculate performance statistics for the complete AGV fleet.

3.3 Transport Tasks

Transport operations in the warehouse are represented as tasks. Each task has a unique identifier and contains a **pickup location** and a **delivery location**. A newly generated task initially has the status **waiting**, meaning that it has not yet been assigned to an AGV.

During its lifetime, a task progresses through several stages. After an AGV is selected, the task becomes assigned. When the robot reaches the pickup location, the item is collected, and after the AGV reaches the delivery location, the task is marked as completed.

To support performance evaluation, four timestamps are stored for every task: the time at which it was created, assigned to a robot, picked up, and completed. These timestamps make it possible to calculate important experimental metrics. For example, **queue time** is calculated as the difference between assignment and creation time, while **cycle time** represents the total time between task creation and completion.

3.4 Task Assignment

The task assignment component determines **which available AGV should receive each waiting transport task**. The dispatcher considers only robots that do not currently have an assigned task, meaning that a robot must be available before it can receive a new transport request.

Two dispatch strategies are implemented in the simulator: **nearest-available** and **first-available**. These strategies are intentionally simple so that their effect on warehouse performance can be compared experimentally.

In the **nearest-available** strategy, the dispatcher estimates the path from every available robot to the task's pickup location using A* pathfinding. The robot with the shortest reachable path is selected. This approach attempts to reduce unnecessary travel before pickup by assigning a task to the robot that is currently closest in terms of route length.

The **first-available** strategy acts as a simpler baseline. Instead of comparing the distances of all available robots, it selects the first available robot that can reach the pickup location. This strategy requires less decision-making but does not attempt to minimize the distance travelled before collecting the item.

After a robot has been selected, the task is assigned to that AGV and its assignment timestamp is recorded. The robot then changes to the **to_pickup** state so that the route planner can generate a path towards the pickup location.

The two dispatch strategies are later compared experimentally to determine whether selecting the nearest available AGV improves performance compared with the simpler first-available approach.

3.5 Path Planning

Path planning determines how an AGV moves from its current position to a required destination while respecting the warehouse layout. The simulator uses two related forms of the A* algorithm: standard A* for estimating routes during task assignment and Space-Time A* for coordinated movement of multiple AGVs.

Standard A* is used by the dispatcher when comparing available robots. For each candidate AGV, a route to the pickup location is calculated, and the resulting path length can be used to determine which robot is closest to the task.

For actual multi-robot movement, the simulator uses **Space-Time A***. Unlike ordinary path planning, which represents a state only by a grid position, Space-Time A* represents each state as a combination of **position and simulation time**. This allows the planner to consider not only whether a cell is physically accessible, but also whether another AGV is expected to occupy that cell at a particular time.

The algorithm uses the Manhattan distance as its heuristic, which is suitable for the warehouse because AGVs can move only horizontally or vertically. The search evaluates possible neighbouring cells and chooses routes according to the travelled cost combined with the estimated remaining distance to the destination.

In addition to normal movement, Space-Time A* allows an AGV to **wait in its current position for one simulation step**. Waiting can be necessary when another robot has reserved a cell that would otherwise be part of the shortest route. Therefore, a safe path may contain the same grid position in two consecutive time steps.

The goal of each robot depends on its current operating state. A robot travelling to collect an item uses the task's pickup point as its goal, while a robot carrying an item uses the delivery point. A robot returning after completing its work uses its home position as the destination.

For multi-AGV operation, routes are planned in a coordinated manner rather than independently. The planner generates paths for the robots within a defined planning horizon and uses information about previously planned robot movements when calculating subsequent routes.

The details of how these planned movements are reserved and how conflicts between AGVs are prevented are described in the next section, **3.6 Collision Avoidance and Reservations**.

3.6 Collision Avoidance and Reservations

Because several AGVs operate in the same warehouse, individually valid paths may still conflict with each other. To prevent this, the simulator uses a reservation table together with Space-Time A*. The reservation table stores both occupied cells and movements between cells at specific simulation times. This is a form of cooperative multi-agent pathfinding, where routes are planned in space and time while respecting the planned motion of other agents. More general MAPF methods such as Conflict-Based Search provide optimal solutions under formal assumptions, but at greater algorithmic complexity.

Two main types of conflict are considered. A **cell conflict** occurs when two robots would occupy the same grid cell at the same time. An **edge conflict** occurs when two AGVs attempt to move through the same connection in opposite directions during the same simulation step, which would cause the robots to swap positions. The Space-Time A* algorithm checks both types of reservation before accepting a movement.

When a safe route is generated for a robot, every position on the route is reserved for the corresponding time step. The movement between consecutive cells is also stored as an edge reservation. The destination can additionally remain reserved briefly after arrival, preventing another AGV from immediately planning a conflicting movement into the same location.

The planner uses a **prioritized planning** approach. Robots are considered in sequence, and after one AGV receives a safe path, that path is added to the reservation table. The next robot must therefore generate its path while respecting the reservations created by robots planned earlier. Robots without a current movement goal are treated as stationary, and their current cells are reserved so that moving AGVs cannot plan through them.

To avoid always giving the same AGV the highest planning priority, the order of the robots can be rotated between replanning operations. This distributes planning priority more evenly across the fleet.

If Space-Time A* cannot find a safe route for a robot within the current planning horizon, that robot is temporarily treated as stationary. The planner then recalculates the coordinated routes so that the remaining AGVs plan around its position rather than following potentially conflicting paths.

As an additional safety mechanism, the simulator checks robot positions after movement. It detects both cases where two AGVs occupy the same cell and cases where two robots swap positions during one movement step. If such a conflict occurs, the simulator raises an error. This check is used as a validation mechanism for the coordinated planner rather than as the primary collision-avoidance method.

3.7 Simulation Process

The simulation is executed as a sequence of discrete time steps. At each step, the system updates task generation, robot movement, task states, task assignment, and route planning. This structure ensures that all AGVs operate according to the same simulated timeline.

New transport tasks are generated at fixed intervals defined in the simulation configuration. For each new task, the pickup and delivery locations are selected from the predefined warehouse locations using a seeded random generator. The random seed makes the experiment reproducible, because the same configuration and seed produce the same sequence of generated tasks. Task generation stops once the official simulation duration has been reached.

After task generation, the AGVs execute one step of their previously planned routes. The paths have already been coordinated by Space-Time A*, so the robots can follow them one position at a time. The simulation then checks the new positions to verify that no same-cell or swap collision has occurred.

The simulator then checks whether any robot has reached an important task location. When an AGV reaches a pickup point, the corresponding task is marked as picked up and the robot's next goal becomes the delivery location. When the delivery point is reached, the task is marked as completed and the robot releases the transported item. After completing a task, the AGV can return to its home position before becoming fully idle again.

Waiting tasks are then passed to the dispatcher. If an available AGV is selected for a task, the assignment changes the robot's goal and coordinated replanning is required. Replanning is also triggered when important robot states change, such as reaching a pickup or delivery location, or when a robot has no usable route to its current goal.

The planner recalculates coordinated routes for the fleet using Space-Time A* and the reservation table. A limited planning horizon is used, and planning priority can rotate between AGVs during successive replanning operations. This allows the simulator to respond to changing task assignments and robot goals while maintaining collision-free movement.

The official experiment duration is **600 simulated seconds**. After this point, no new tasks are created, but the simulation is allowed to continue in a **draining phase** so that tasks already generated can still be completed. The simulation ends when all tasks have been completed and no robot has an active transport task. A maximum drain time is also defined as a safety limit to prevent an experiment from continuing indefinitely.

When the simulation finishes, the final robot and task data are converted into a `SimulationResult`. This result contains the performance measurements required for later analysis, including throughput, queue time, cycle time, travelled distance, fleet utilization, completion rate, and drain time.

4. Experimental Methodology

4.1 Experimental Variables

The experimental evaluation was designed to investigate how the **number of AGVs**, **task-generation rate**, and **dispatch strategy** affect warehouse performance. Each experimental configuration was executed several times using different random seeds.

The main experimental parameters were:

Parameter	Values
Number of AGVs	1, 2, 3, 4, 5, 6
Task-generation interval	10 s, 15 s, 20 s
Dispatch strategy	nearest, first-available
Random seeds	1, 2, 3, 4, 5
Official simulation duration	600 simulated seconds
Maximum drain time	1200 simulated seconds

The complete experiment therefore contained **180 individual simulation runs**:

6 fleet sizes × 3 task intervals × 2 dispatch strategies × 5 random seeds = 180 experiments.

The warehouse layout, pickup and delivery locations, movement rules, and path-planning mechanism remained unchanged between experiments. This allowed the effects of fleet size, workload, and dispatch strategy to be studied independently.

4.2 Workload Scenarios

Three task-generation intervals were selected to represent different warehouse workloads. A shorter interval generates transport requests more frequently and therefore places greater demand on the AGV fleet.

A task interval of **10 seconds** represents the highest tested workload. A new transport request is generated approximately every 10 simulated seconds. The **15-second** interval represents a medium workload, while the **20-second** interval represents the lowest tested workload.

Using several workload levels makes it possible to determine whether additional AGVs are beneficial under both congested and lightly loaded conditions. It also makes it possible to identify situations in which increasing the number of robots no longer produces a substantial performance improvement.

4.3 Dispatch Strategies

Two task-assignment strategies were compared during the experiments.

The **nearest-available** strategy evaluates the available AGVs and assigns a task to the robot with the shortest calculated path to the pickup location. The purpose of this strategy is to reduce unnecessary travel before the robot collects the requested item.

The **first-available** strategy selects the first available robot that is able to reach the pickup location. It does not compare the travel distances of all available robots and therefore serves as a simpler baseline strategy.

All combinations of fleet size and workload were tested using both strategies. This allows the results to show whether the additional distance-based decision made by the nearest-available strategy provides a measurable performance benefit.

4.4 Performance Metrics

Several performance metrics were collected from every simulation run.

Throughput represents the number of transport tasks completed per minute during the official 600-second experiment period. Only tasks completed before or at the experiment horizon are included in this calculation.

Average queue time measures how long a task waits before being assigned to an AGV. It is calculated as the difference between the task assignment time and task creation time.

Average cycle time measures the total time required to process a transport request, from the moment the task is created until its delivery is completed.

Fleet utilization represents the percentage of the available AGV fleet time during which robots are assigned to transport tasks. This metric provides an indication of how heavily the available robot capacity is being used.

Completion rate represents the percentage of generated tasks that are completed within the official 600-second simulation horizon.

Total travel distance is the total number of grid cells travelled by all AGVs during an experiment. This metric is particularly useful when comparing dispatch strategies because a more efficient assignment policy may reduce unnecessary robot movement.

Drain time measures the additional simulated time required after the 600-second experiment horizon to complete tasks that were already generated but had not yet finished.

These metrics provide different perspectives on system performance. Throughput and completion rate measure productivity, queue and cycle time measure service quality, utilization indicates the use of robot capacity, and travel distance provides an indication of movement efficiency.

4.5 Reproducibility

Randomness is used when selecting pickup and delivery locations for newly generated tasks. To make the experiments reproducible, each simulation uses a predefined **random seed**.

Five seeds, numbered from 1 to 5, were used for every experimental configuration. Results from these repetitions were then combined by calculating the **mean** and **standard deviation** of the measured performance metrics.

The same seeds were used when comparing different fleet sizes and dispatch strategies. This reduces the possibility that differences between configurations are caused only by different randomly generated task sequences.

The experiments were executed in **headless mode**, without the graphical Pygame visualization. In this mode, simulated movement follows the same logical rules as the visual simulator but is executed as quickly as possible, making it practical to run the complete set of experiments.

4.6 Simulation Horizon and Draining

Task generation is stopped when the official simulation time reaches **600 seconds**. However, immediately stopping the simulator at this point could leave tasks unfinished, especially under high workloads or with small AGV fleets.

For this reason, the simulator enters a **draining phase** after the official experiment horizon. During this phase, no new tasks are generated, but the AGVs continue processing tasks that already exist.

The simulation finishes when all generated tasks have been completed and no AGV has an active transport task. A maximum drain period of **1200 additional simulated seconds** is used as a safety limit to prevent a simulation from running indefinitely.

Performance measures such as throughput and completion rate are calculated using the official 600-second horizon so that additional drain time does not artificially improve the measured productivity of a configuration. Drain time is instead recorded separately as an additional performance metric.

4.7 Analysis Method and Validation

For each combination of fleet size, workload, and dispatch strategy, the five random-seed runs were aggregated using the arithmetic mean and sample standard deviation. The error bars in Section 5 therefore show mean $\pm$ one standard deviation. The analysis is descriptive rather than inferential: the experiment is used to identify engineering trends and operating thresholds, not to claim statistical significance from a large population sample.

The implementation was checked with an automated pytest suite containing 10 tests. These tests cover idle and parking behaviour, path-start validation, cell and edge reservations, temporary destination holding, Space-Time A* waiting, prevention of direct position swaps, blocked-route behaviour, and coordinated multi-robot planning. The final dataset was also validated: all 180 planned runs were present, every configuration had exactly five seeds, there were no duplicate configurations, all generated tasks were eventually completed, and no run ended with waiting tasks. The largest observed drain time was 1199 s, below the configured 1200 s drain limit.

5. Results and Discussion

This section combines numerical results with their engineering interpretation. Unless otherwise stated, fleet-size results use the nearest-available dispatch strategy. Every plotted point is the mean of five seeds and the error bars show one standard deviation. The emphasis is on identifying capacity thresholds, diminishing returns, and the effect of dispatch policy.

5.1 Fleet Capacity, Throughput, and Utilization

The throughput and utilization results reveal a clear transition from capacity-limited to demand-limited operation. With one AGV, throughput is approximately 2.12 tasks/min for all three workloads and utilization is 100%, showing that the robot is saturated. Under the 10-second workload, throughput rises to 4.08 tasks/min with two AGVs and 6.04 tasks/min with three, after which it remains close to 6.1 tasks/min. The 15-second workload reaches 4.08 tasks/min with two AGVs and then approaches a plateau near 4.2 tasks/min, while the 20-second workload reaches its plateau of about 3.26 tasks/min with two AGVs.

Utilization supports the same interpretation. For the highest workload, three AGVs remain 98.70% utilized, but utilization falls to 67.62% with four and 41.19% with six. For lighter workloads the decline begins earlier. Additional AGVs therefore provide large gains only while fleet capacity is below incoming demand; after the throughput plateau is reached, extra robots mainly create spare capacity.

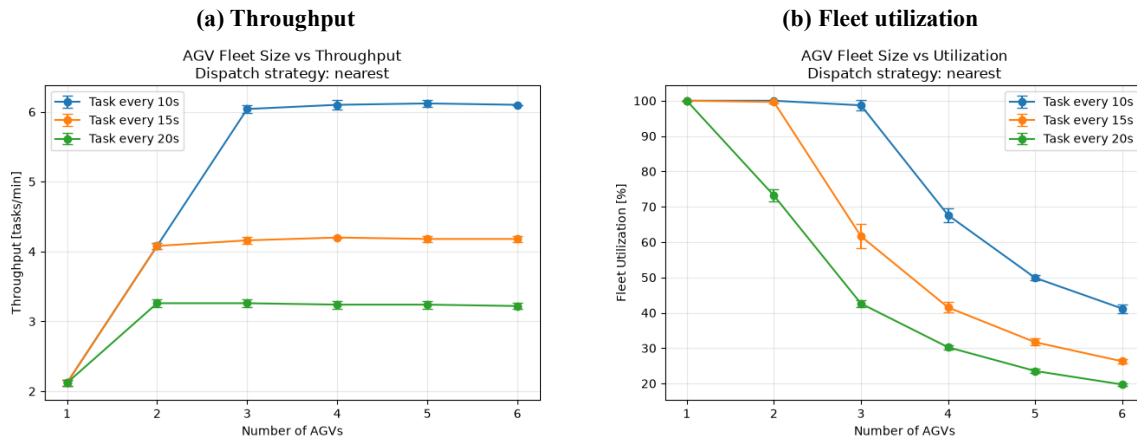


Figure 2. Fleet-size effect on (a) throughput and (b) fleet utilization for the nearest-available strategy.

5.2 Delay, Completion, and Backlog

Delay metrics are more sensitive than throughput to an undersized fleet. At the 10-second workload, average queue time falls from 577.17 s with one AGV to 155.03 s with two, 10.54 s with three, and 1.15 s with four. Average cycle time follows the same pattern, decreasing from 605.01 s to 183.55 s, 39.35 s, and 27.34 s respectively. For the medium workload, three AGVs reduce queue time to 1.96 s; for the low workload, two AGVs already reduce it to 5.21 s.

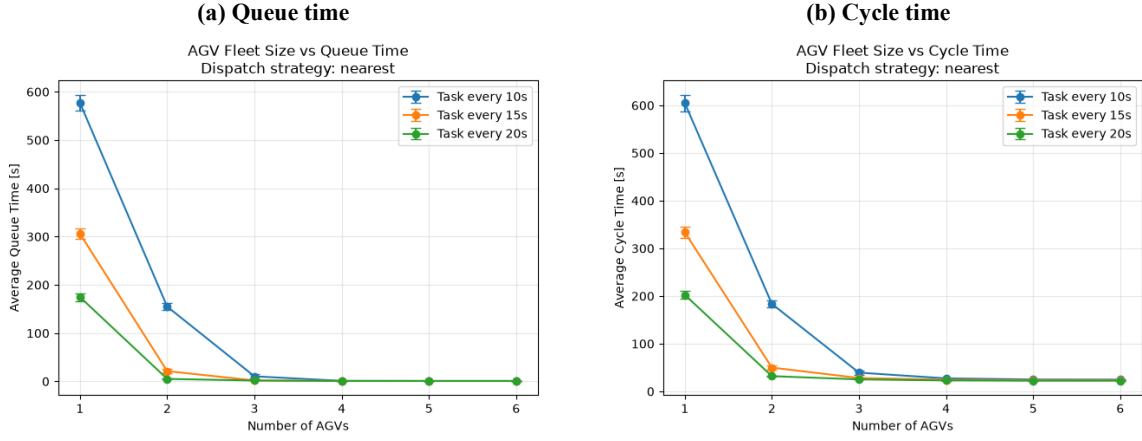


Figure 3. Fleet-size effect on (a) average queue time and (b) average cycle time.

Completion rate and drain time show whether a configuration can keep up with demand within the official 600-second horizon. With one AGV, completion is only 33.65%, 49.30%, and 64.25% for the 10-, 15-, and 20-second workloads. Under the highest workload, three AGVs raise completion to 95.88% and reduce average drain time from 1153.4 s to 22.4 s. Under the medium workload, two AGVs reach 94.88% completion but still have a 21.34 s queue; a third AGV reduces that queue to 1.96 s. Under the low workload, two AGVs reach 98.79% completion with only 6.2 s of drain time.

These results show why throughput alone is insufficient for fleet sizing. A configuration can be close to the maximum throughput while still producing non-trivial queueing or incomplete work at the experiment horizon. In engineering terms, the preferred fleet size therefore depends on whether the objective is minimum capital capacity or a stricter service-delay target.

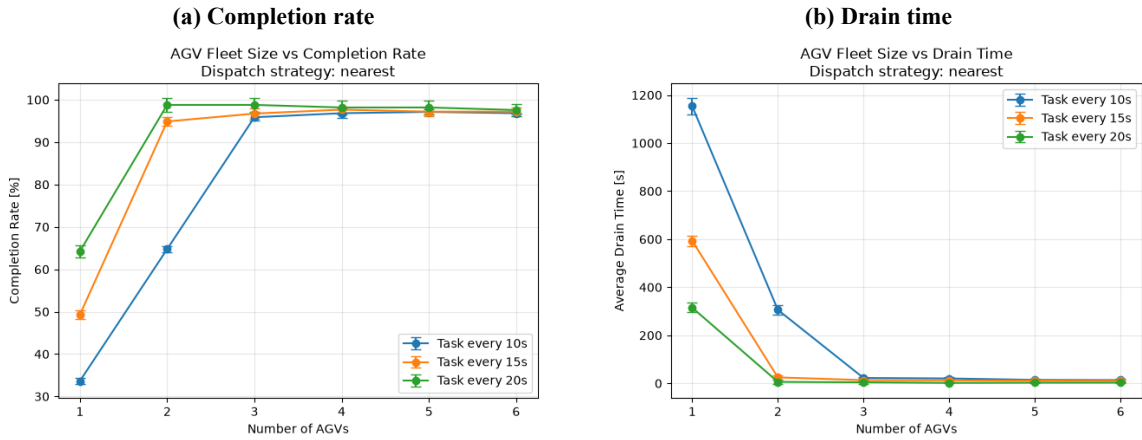


Figure 4. Fleet-size effect on (a) completion within the 600-second horizon and (b) post-horizon drain time.

5.3 Dispatch Strategy Comparison

The nearest-available and first-available strategies are almost identical for fleets of one to three AGVs under the 10-second workload. In this region, capacity is tight and the dispatcher has little meaningful choice between idle robots. Differences appear once four or more AGVs are available. With six AGVs, nearest-available reduces average cycle time from 28.28 s to 24.76 s, a reduction of about 12.4%.

The effect is clearer in total travel distance. Relative to first-available, nearest-available reduces mean distance by about 7.1% with four AGVs, 10.5% with five, and 12.6% with six. Throughput remains almost unchanged between the strategies. The dispatch policy therefore primarily improves movement efficiency once the fleet has enough spare capacity for assignment choice, while overall system capacity remains controlled mainly by fleet size and task arrival rate.

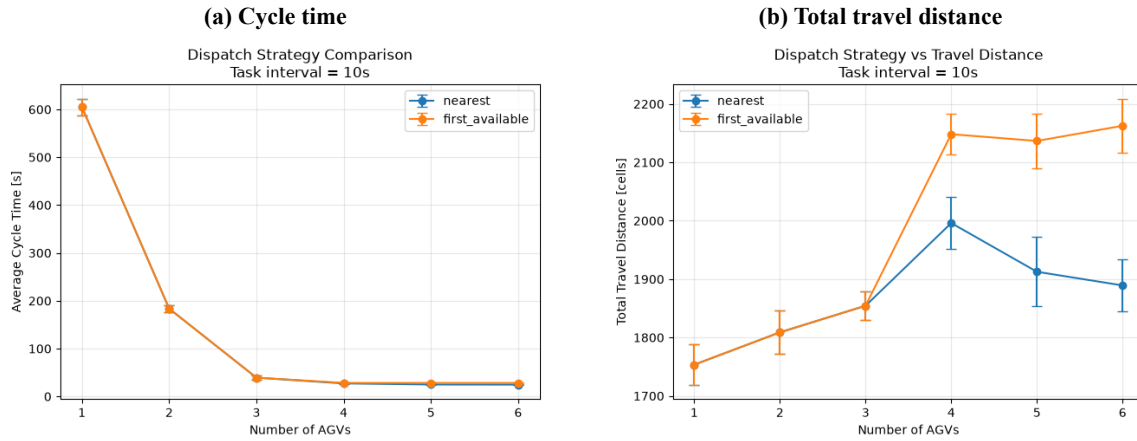


Figure 5. Dispatch-strategy comparison at the 10-second workload: (a) average cycle time and (b) total AGV travel distance.

5.4 Answers to the Research Questions

The first research question asked how fleet size and task arrival rate affect warehouse performance. The evidence shows that the required fleet size increases with workload, but only up to a clear capacity threshold. The second research question asked whether nearest-available dispatching improves efficiency. The answer is yes for larger fleets, especially in travel distance, but the improvement is smaller than the effect of adding enough AGVs to satisfy demand.

Workload	Observed capacity point	Throughput	Queue time	Interpretation
High: 10 s	3 AGVs	6.04 tasks/min	10.54 s	Three AGVs reach demand capacity; a fourth mainly improves service delay.
Medium: 15 s	2-3 AGVs	4.08-4.16 tasks/min	21.34 $\rightarrow$ 1.96 s	Two AGVs provide most throughput; the third removes most remaining queueing.
Low: 20 s	2 AGVs	3.26 tasks/min	5.21 s	Two AGVs are sufficient for almost all generated demand.

Table 2. Observed fleet-capacity points for the nearest-available strategy. These are engineering operating points from the tested scenarios, not universal fleet-size rules.

6. Threats to Validity and Limitations

Internal validity is strengthened by fixed experimental conditions, seeded task generation, repeated runs, automated tests, and runtime collision assertions. However, each configuration uses only five random seeds and the analysis is descriptive; no formal hypothesis tests or confidence intervals are used. Small differences between nearby configurations should therefore be interpreted cautiously, especially when error bars overlap.

Construct validity is limited by the simplified performance model. Fleet utilization measures the proportion of available robot time during which AGVs are assigned to transport tasks, not the percentage of time physically moving. Queue and cycle time are measured in discrete simulated seconds, and loading/unloading duration, turning cost, acceleration, and variable speed are not represented.

External validity is limited by the fixed 15 x 20 warehouse layout, two pickup points, two delivery points, and regular task-generation intervals. Real warehouses may contain more stations, dynamic demand, task priorities, narrow one-way aisles, temporary obstacles, battery constraints, charging, and heterogeneous vehicles. The numerical fleet thresholds reported here therefore apply to the tested model rather than to warehouses in general.

Algorithmically, prioritized Space-Time A* provides safe coordinated paths for the tested scenarios but does not guarantee a globally optimal multi-agent solution. Planning also uses a finite horizon and

only two dispatch strategies were compared. More advanced MAPF and dispatch methods could produce different trade-offs, particularly in denser layouts or much larger fleets.

7. Conclusion

This project developed a reproducible multi-AGV warehouse simulation that integrates task generation, two dispatch strategies, A* route estimation, Space-Time A* planning, reservations, replanning, parking behaviour, and collision validation. The simulator was evaluated over 180 controlled runs using six fleet sizes, three workloads, two dispatch strategies, and five random seeds.

The results show that fleet sizing is the dominant determinant of throughput and delay. One AGV is saturated in every tested workload. The high workload requires about three AGVs to reach demand capacity and a fourth if near-zero queueing is desired; the medium workload requires roughly two to three AGVs; and the low workload is effectively served by two. Beyond these points, throughput plateaus while fleet utilization falls, demonstrating diminishing returns from additional robots.

Nearest-available dispatching does not materially change throughput, but it becomes valuable when larger fleets provide a genuine choice of available robots. Under the highest workload it reduces total travel distance by approximately 7-13% for fleets of four to six AGVs and produces a smaller reduction in cycle time. The main engineering conclusion is therefore that fleet capacity should first be matched to demand, after which dispatch policy can be used to improve movement efficiency.